جامعة خليفة
Khalifa University

WASTE MANAGEMENT

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1. Purpose:

The purpose is to develop, implement and maintain a procedure to manage generated waste to achieve the best EHS performance with regard to Management System and the communal and economic development of Abu Dhabi Emirate as a whole.

This is to ensure that all waste materials are;

- Properly categorized for correct collection, storage, handling, treatment and disposal
- Managed in accordance with the relevant UAE regulations and standards
- Minimized, reduced, reused and recycled as far as possible.
- The procedure is meeting the requirements of the Abu Dhabi EHSMS, Federal Law No. (24) 1999, and Abu Dhabi Law No. (21) 2005, related to "Waste Management". Law No. (16) of 2005 pertaining to the Reorganization of the Abu Dhabi Environment Agency.

This procedure is for all laboratories consume materials that are hazardous. Such use generates waste that is hazardous. The majority of the waste resulting from laboratory operations is not hazardous, but that which is hazardous can injure the people who must handle or transport it. Laboratory waste can contaminate sewer systems and other treatment facilities, and it can pollute the environment.

This procedure is also meant to assist in the development of waste management strategies, based on the principle of hierarchical control, and promote the sustainable development. This procedure aims to minimize the waste generation, reduce the natural resource consumption and environmental impact arising from waste disposal.

2. Scope:

This procedure applies to KU operations and services regarding the management of all hazardous and non-hazardous wastes and to all waste generation activities (in the form of liquids and solids).

This procedure applies also to waste discharged to the atmosphere in the form of gases, vapors, fumes, aerosols, dusts and particulates, or to radioactive wastes.

3. Responsibilities:

3.1 EHS Steering Committee Representatives: Responsible for

- Ensure necessary resources are available for the effective implementation and maintenance of this procedure.

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- Reporting to top management on the performance of this procedure and any need for improvement.

3.2 Department Directors: Responsible for

- Ensure implementation and monitoring of this procedure effectively.

3.3 EHS Manager: Responsible for

- Ensure proper coordination with EHS Management Committee representatives with all sections for consolidation and disposal of Waste Management activities within KU.
- Overall responsibility for implementing this procedure effectively and communicating this procedure to all levels of KU.
- Ensure compliance with the relevant Environmental Health and Safety Regulation / objectives and targets and other legal requirements, EHS Management System Policy and Procedures. Ensure compliance to the requirements of statutory and regulatory, international and local, and KU requirements related to waste management.
- Facilitate EHS improvement advices relevant to this procedure.
- Provide necessary consultancy for all concerned for effective implementation of this procedure.
- Report all incidents related to waste management to Abu Dhabi Education Council.

3.4 Facilities Management Department: Responsible for

- Implementing this procedure effectual.
- Ensuring necessary corrective and preventive actions.
- Ensuring the waste management issues are addressed as appropriate.
- Reviewing the lessons learned from the implementation and improves it.
- Identifying, evaluating the generated waste, and proper waste management within Departments / Sections.

3.5 Laboratory Managers: Responsibilities

- Laboratory Managers are officially responsible for proper waste management.
- Laboratory managers must be familiar with the regulations and standards that apply to their particular organizations.
- Managers must design a system for preventing pollution, as well as for the safe handling of unavoidable waste that provides for proper collection, segregation, storage, disposal, monitoring, quality control, and record keeping.

- Laboratory management has the responsibility to train employees in the use of engineering controls and personal protective devices, and to ensure that its employees comply with safety directives.
- Some laboratories find it more efficient to designate a laboratory waste Coordinator to maintain records, keep emergency call lists current, and oversee the laboratory's waste management program.

3.6 EHS Section Coordinator: Responsible for

- For effective implementation of this procedure in all sections in KU.
- Preparing waste management registers and periodical reporting.
- Coordinating with approved waste management contractors.
- Carry out incident investigation related to waste management.

3.7 EHS Inspectors: Responsible for

- Inspection / monitor all waste management activities.
- Report all non-conformities all incidents related to waste management.

3.8 Employee Responsibilities:

Employees must comply with established procedures. Within the organization, technical employees have the best opportunity to identify those wastes that need special handling. Bench technologists are responsible for segregating and labeling all waste that requires special handling at the point of waste generation. Employees must bring to management's attention unsafe working conditions and identify opportunities for hazardous waste reduction.

3.9 Waste Contractor: Responsibilities

- Waste shippers and contractors have legal and contractual obligations of proper waste management.
- Because environmental laws are complex and specialized, many laboratories seek compliance assistance advice from their waste contractors.

4. Abbreviation / Definitions:

MSDS: Material Safety data Sheet is a form provided by the supplier of chemical product / manufacturer which containing data regarding the properties of a particular substance. It's intended to provide workers and emergency personnel with procedures for handling or working with that substance in a safe manner.

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Wastes: An unwanted or unusable substance or Material. Wastes are objects which are disposed or are intended to be disposed or are required to be disposed of by the provisions of national laws.

Solid Waste: Non-liquid, non-soluble materials ranging from municipal garbage to industrial wastes that contain complex and sometime hazardous substances. Or any discarded material that is abandoned by being disposed of, burned or incinerated, recycled For example, domestic, industrial, agricultural, medical, construction and demolition wastes and maintenance consumables.

Waste Producer: A person or organization whose action gives rise to hazardous and / or general waste as a result of its activities.

Waste Management: Segregation, collection, storage, and disposal through Central of Waste Management–Abu Dhabi registered waste handling contractor.

Solid Waste Management: Supervised handling of waste materials from their source through recovery processes for disposal.

Waste Hierarchy: Principle of managing waste which recognizes the benefits of reducing, reusing, and recycling wastes as preferred options over disposal.

Waste Treatment: Physical, chemical or biological processing of waste in order to make it harmless, reduce its volume or for recycling.

Waste Handling: All operations from the point of generation of waste to the safe disposal, including collection, storage, treatment and recycling or disposal.

Waste Reduction: Process of reducing the quantity of waste produced through the review of operational practices and optimal use of raw materials.

Waste Recycling: The reprocessing of waste into the same or a different product. Typical recyclable wastes include glass, paper, plastics etc.

Waste Disposal: Operations not involving extraction or re-use of substances, such as burial, deep injection, biological or physical, chemical treatment, permanent storage, destruction or any other method approved by the competent authorities.

Substance: Any chemical element and its compounds and biological entity or microorganism, except radioactive substances or genetically modified organisms.

Waste Regulation Authority: The authority charged with the co-ordination and control of activities of waste producers, disposal facilities and contractors regarding the management of waste and ensuring that these activities conform to the relevant legislation.

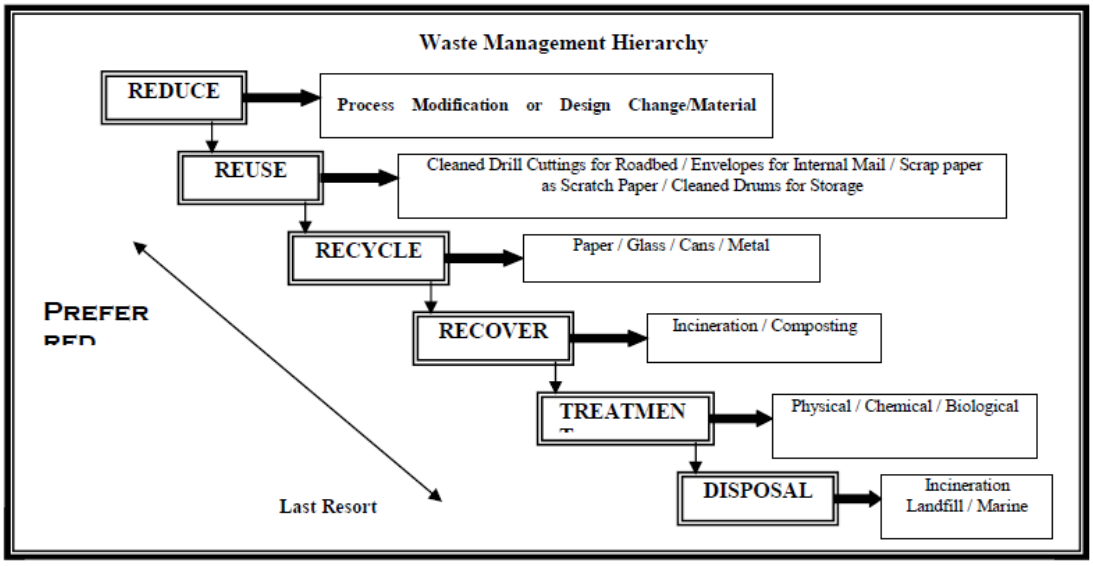
Waste Contractor: A person or organization, who controls the transportation, temporary storage, and final treatment / disposal of waste once it has left the KU and approved as appropriate.

Sustainable development: Development which can meet the needs of the present, without compromising the ability of future generations to meet their needs.

5. Procedure:

5.1 Waste Generation:

All practical measures to reduce the generation of waste and to recover valuable resources must be given due consideration by all operations to the use of the waste management hierarchy.



5.2 Waste Identification:

Waste identification and classification are undertaken by KU sections through their concerned EHSMS coordinators for an effective management of waste, taking into account the following criteria;

- Hazardous waste.
- Recyclable waste.
- General waste.

5.3 Hazardous waste:

Hazardous wastes have one or more of the characteristics of ignitibility, corrosively, reactivity, toxicity, and un-degradability.

This includes waste such as:

- Laboratory Wastes, e.g. bio Wastes, radioactive materials and chemicals.
- Filter cartridges & carbon filters.
- Battery Acids.
- Electrical waste.
- Lubricating Oils.
- Materials or packaging of materials classed as “harmful to the environment”.
- E- Waste (printer cartridges, PC monitors, etc.).
- Equipment scrap.

5.4 Non-Hazardous Waste:

The non-hazardous waste is a waste which does not have the characteristics of the hazardous waste such as;

- Paper and cardboard.
- Wood.
- Scrap-metal (no hydrocarbons).
- Food (from food outlets).

It was important to identify the waste classification at the time of production to prevent the need for further segregation at later stages. Subsequent holders of the waste must be provided with a description, which is precise enough to enable them to avoid mismanaging the waste. The description should always mention any special hazards associated with the waste, precautions, training, and knowledge requirements for handling the waste. It also includes if possible the name of the substance(s) contained within

the waste and includes details of safe handling procedures as set in the waste declaration form.
Food wastes and associated garbage which may carry disease or pests should be kept separate from the garbage which does not contain such food wastes. The waste storage containers are of an appropriate type to ensure the safe storage of waste while taking into account the above mentioned precautions. The wastes, which are returned to suppliers, are stored separately.

5.5 Waste Classification and Segregation:

A waste is any material or substance (Solid/semisolid/Liquid/Gaseous) that has been discarded or intended to be discarded.
A waste is generally defined by its physical, biological and chemical characteristics.

There are five generic waste types:

a) Municipal Waste
b) Non-Hazardous Industrial Waste
c) Hazardous Waste
d) Radioactive (A special category of hazardous waste)
e) Medical Waste (A special category of hazardous waste)

A. Municipal Waste:

Inert or non-hazardous waste produced through everyday non-industrial operations and living. Examples are metal tin/containers, kitchen wastes etc.

- **Food waste**, non-edible by-products and other refuse shall be separate and removed from rooms where food is present as quickly as possible to avoid their accumulation and any risk of cross-contamination.
- Food waste, non-edible by-products and other refuse shall be deposited in closable containers and disposed of appropriately. Containers shall be constructed of suitable leak-proof, impervious material that is easy to clean or disposable.
- Containers used to hold dangerous substances should be labeled and, where appropriate, be lockable to prevent malicious or accidental contamination of food.
- All waste shall be eliminated in a hygienic and environmentally friendly way.

- The greasy waste should be promptly removed and the adjoining floor surfaces should be cleaned thoroughly afterwards.

B. Non-Hazardous Industrial Waste:

Any waste generated by industrial processes or civil activities that is not classified as hazardous. Examples are masonry and brick rubble, wood and metal etc.

C. Hazardous Waste:

Substances that pose a present or future threat to man and/or the environment which are unwanted and economically unusable and which exhibit one or more of the following characteristics: Ignitable, Corrosive, Reactive or Toxic

D. Radioactive

Any waste containing radionuclide. (i.e. A nuclide that is radioactive)

E. Medical Waste:

Any waste which is generated in the diagnosis, treatment, or immunization of human beings, in related research, biological productions, or testing.

- **Hazardous Waste Classifications:**

  Due to potential impacts of hazardous wastes and their variety in nature, specific classifications are required so proper waste management practices can be applied.

  Materials/substances intended to be discarded that are identified in Table 1 below are hazardous, unless proven otherwise.

Table 1: List of Known Hazardous Wastes from Oil & Gas Industry Operations:

Acid sludge	Laboratory Chemicals
Ammonia	Lead based pipe compounds
API separator sludge	Light bulbs/tubes
Asbestos	Medical Waste
Batteries	NORM or LSA scales (Radioactive)
Biocides/Pesticides/Herbicides	Paint and thinners
Catalysts	PCB's
CFC's/Halons (Contained)	Photographic chemicals
Contaminated absorbents	Pigging wastes
Coolants	Pit sludge
Crude oil/condensate	Pyrophoric scale

Crude contaminated soils	Radioactive sources
Distillation residues and condensate	Solvents
Effluent treatment sludge	Spent activated carbon
Explosives	Tank bottom sludge
Filter cake (mercury contaminated)	Tar
Filters (e.g. Oil, condensate, glycol, amine)	Treatment Chemicals
Hydraulic fluids	Used lubricants
Insulation Material (Certain Types)	

- **Types of Laboratory Chemical Waste:**
  As a first step in any program to properly manage laboratory wastes, an institution should carefully survey its waste generation activities and identify those wastes that must be managed, handled, and disposed of separately.

  Four chemical characteristics are used by the EPA to identify hazardous chemical waste: ignitability, corrosivity, reactivity, and toxicity. Of course, chemical waste can have multiple characteristics, as well as the characteristics of radioactivity and infectiousness.

  A material/ substance to be discarded that is not included in Table 1 is classified as hazardous waste if it exhibits one or more of the following characteristics:

  a) Ignitability
  b) Corrosivity
  c) Reactivity
  d) Toxicity

A. Ignitability:
A waste is ignitable if:

- It is a liquid and has a flash point less than 21°C.
- It is not a liquid and is capable of causing fire through friction, absorption of moisture, or spontaneous chemical changes; and, when ignited, burns so vigorously and persistently that is creating a hazard.
- It is an ignitable compressed gas or
- It is an oxidizer.

Table 2: Examples of Ignitable Chemical Waste:

Flammable Liquids	Flammable Gases	Oxidizers
Organic solvents such as	Hydrogen, butane	Nitrate salts,

Acetone, alcohols, toluene, Xylene		peroxides

B. Corrosivity

Liquid, corrosive hazardous waste has a pH ≤2 or ≥12.5; or it has the capacity to corrode steel. Examples include mineral acids, such as sulfuric, hydrochloric, and phosphoric acid, or bases such as ammonium hydroxide. D002 is the identification number of corrosive waste.

C. Reactivity

A waste is reactive if:

- It reacts violently with water.
- It forms potentially explosive mixtures with water.
- It is normally unstable and readily undergoes violent change without detonation.
- When mixed with water; it generates toxic gases, vapors, or fumes in a quantity sufficient to present a danger to human health or the environment.
- It is readily capable of detonation or explosive decomposition under normal conditions (standard temperature and pressure).

Table 3: Examples of Reactive or Potentially Reactive Waste

Reactive	Potentially Explosive	Toxic Gas Source
Sodium, potassium, and other Alkali metals	Dry picric acid; ether that Contains peroxides	Cyanide or sulfide solution

D. Toxicity

A waste is toxic if laboratory testing indicates that it contains a contaminant listed in Table No. 4 (List of Toxic Substances) at a concentration equal to or greater than (≥) the maximum acceptable concentration level given.

Table 4: List of Toxic Substances*:

Containment/Substance	mg/ltr	Containment/Substance	mg/ltr
1,1-Dichloroethylene*	0.70	Hexachloroethane*	3.00
1,2-Dichloroethylene*	0.50	Hydrocarbons	(d)
1,4-Dichlorobenzene*	7.50	Inorganic cyanides*	(c)
2,4-D*	(b)	Inorganic halogen-containing compounds*	(c)
2,4-Dinitrotoluene*	0.13	Inorganic sulfer-containing compounds*	(c)

2,4,5-TP (Silvex)*	(b)	Lead (Tetraethyl-lead*)	5.00
2,4,5-Trichlorophenol	400.00	Lindane*	0.40
2,4,6-Trichlorophenol	2.00	Mercury	0.20
Antimony and antimony compounds	0.005	Metal Carbonyls*	0.40
Arsenic and arsenic compounds	5.00	Methoxychlor	10.00
Barium and barium compounds	100.00	Methyl ethyl ketone	200.00
Benzene*	0.50	Nickel compounds	(c/e)
Beryllium and beryllium compounds	0.0002	Nitrobenzene	2.00
Boron compounds	0.003	Pentachlorobenzene*	(b)
Cadmium and cadmium compounds	1.00	Pentachloroethlylene*	(b)
Carbon tetrachloride	0.50	Pentachlorophenol*	(b)
Chlordane*	(b)	Phenol and phenol compounds*	19.00
Chlorobenzene	100.00	Pyridine*	5.00
Chloroform	6.00	Selenium	1.00
Chromium (Hexavalent compounds)	5.00	Silver	5.00
Cobalt compounds	0.05	Tellurium and Tellurium compounds	0.10
Copper compounds	0.02	Tetrachlorobenzene*	(b)
Cresol	200.00	Tetrachloroethylene*	(b)
m-Cresol	200.00	Tetrachlorophenol*	(b)
o-Cresol	200.00	Thallium and Thallium compounds	0.10
p-Cresol	200.00	Tin compounds	2.00
Endrin*	(b)	Toxaphene	0.50
Heptachlor (and its epoxide)*	(b)	Trichlorethylene	0.50
Heterocyclic organic compounds*	(c)	Vanadium compounds	(c)
Hexachlorbenzene*	0.13	Vinyl chloride	0.20
Hexachlorobutadiene*	0.50	Zinc compounds	(e/f)
(a) Concentrations in mg/l – referenced to US EPA Environmental Guidelines			
b) Pesticide/Fungicide/Herbicide constituent – Human/Environmental toxin. All pesticide/fungicide/herbicide waste must be cleared through HSE department for disposal requirements.			
(c) Consult HSE department – environmental group for proper handling and disposal procedure.			
(d) Laboratory testing required to determine total petroleum hydrocarbon (TPH) and/or volatile organic compounds (VOC's). Toxic/heavy metals testing may also be required.			
(e) US DOT listed			
(f) NFPA listed			
* Denotes acutely toxic substances			

All solid wastes are being classified, collected, segregated and, where practicable, stored in suitable bags or containers for proper disposal. The

liquid wastes are collected in proper containers and stored properly till the time of disposal. Contaminated water and other liquid wastes are collected directly from the source by the department or person producing them immediately to avoid unnecessary accumulation or leakage to sewerage systems. Proper handling equipment is being used.

The wastes must be so segregated and stored, as to prevent a contact or reaction which may result in the release of toxic or flammable gas, heat, or pose any health, environmental pollution, or fire hazard.

The following Hazardous wastes should not be mixed:

- Acids and Alkaline or Metals,
- Oxidizing Agents and Organic Solvents
- Water reactive wastes and other wastes

Table 5: The table below illustrates the effects of some common storage incompatibilities:

Waste Types	Result
acid + alkali	= heat
acid + hypochlorite	= toxic gas
acid + metal	= toxic gas/ flammable gas
oxidizing or reducing agent + organic solvent	= fire
water reactive + most things	= fire
pyrophoric substance + dry + air	= fire

- **Radioactivity:**
  By definition, radioactivity is the spontaneous disintegration of certain atomic nuclei accompanied by the emission of alpha, beta, and/or gamma radiation.
  Hazardous radioactive wastes are materials containing radioactive nuclei possessing specific radioactivity:
  - Equal to or more than 0.37 (≥0.37) Bq/g of Radium for solids.
  - Equal to or more than 0.37 (≥0.37) mBq/g of Radium for liquids.

  Materials and substances possessing low-level specific radioactivity – i.e. less than the above mentioned limits are considered non-hazardous. Properly sealed radioactive waste materials that are

returned to the supplier when exhausted or no longer needed are not considered hazardous waste.

a. **On-site Accumulation and Storage:**
Careful selection, implementation, and supervision of on-site waste management methods will minimize potential radiation safety hazards and can greatly reduce (and sometimes eliminate) the need for off-site disposal.

Proper segregation of wastes reduces unnecessary generation of wastes and maximizes the efficiency of disposal operations. General segregation guidance is presented in Table 6 below.

Table 6: General Radioactive Waste Segregation Groups:

Keep These Radioactive Waste Types	Separate from These
Radioactive wastes	All other types of wastes
Wastes with known contamination	Wastes with unconfirmed contamination
Liquids	Solids
Aqueous liquids	Nonaqueous liquids
Wastes with only short-lived radionuclides	Wastes containing long-lived radionuclides
Wastes disposable on-site	Wastes that must be shipped off-site
Deregulated liquid scintillation vials	Regulated liquid scintillation vials
Radioactive sharps	Nonradioactive sharps

Radionuclides should be separated by half-life whenever possible. This allows for greater use of the decay-in-storage method of waste disposal and aids in reducing the amount of storage space needed and the amount of radioactive waste that requires disposal.

b. **Waste Collection Containers:**

- Solid Wastes: A container that is clearly labeled and separate from general waste should be used for collection of dry, solid waste such as contaminated paper, gloves, etc. In most cases, cardboard containers or step cans with appropriate plastic bag liners are satisfactory for routine collection of contaminated solid wastes.
- Sharps: Radioactive sharps should be collected separately from nonradioactive sharps and placed in appropriately labeled, puncture-resistant sharps containers.

- Liquid Wastes: A capped, plastic container should be used for liquid waste.
- Labels should be placed on all containers and should list:
  - Identification of the laboratory and authorized user of radioactive materials;
  - Date of accumulation;
  - Radionuclides present and the estimated activity of each; and
  - Source and type of waste.
- A warning label with the NRC-prescribed radiation symbol and the phrase “Caution—Radioactive Material” is required.

c. Radioactive Waste Shielding:

Waste collection, handling, and storage areas should be evaluated by the radiation safety officer to determine the necessity of shielding waste containers, as well as the correct shielding. For example, when shielding is necessary for the storage of wastes containing 32P, a beta emitter of 1/2-inch poly (methyl methacrylate), not lead, should be used. If the waste contains gamma emitter lead may be used. If the waste contains both gamma emitters and 32P, lead is placed around the outside of the poly (methyl methacrylate) shield. Unnecessary use of shielding should be avoided, because it tends to become contaminated and adds to the volume of radioactive waste.

d. Radioactive Waste Security:

Radioactive waste must be secured from unauthorized access or removal. Warning signs indicating the presence of radioactive materials should be placed in all areas where radioactive wastes are collected, processed, and stored.

Personnel with access to waste handling and storage areas should be trained in procedures and work duties.

e. General Treatment and Disposal Methods:

Radioactive waste may be treated and disposed of by several methods, including decay-in-storage; release into the sanitary sewer system; dispersal in the atmosphere; or transfer to a waste disposal firm for off-site processing, incineration, or burial.

The treatment process and method of disposal chosen for each type of waste is dependent on several factors, including the following:

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- The waste type (aqueous, gaseous, solid, organic, or animal carcass);
- The radioisotope and its half-life;
- Combustibility;
- Abu Dhabi regulations; and
- Availability of treatment and disposal facilities.

f. On-site Waste Management:

- **Decay-in-Storage (DIS)**
  Decay-in-storage (DIS) is the preferred method of radioactive waste management for all radioactive materials in any form with a half-life of 120 days or less when the licensee has sufficient space to store the volume generated. Wastes are usually held for a period of ten half-lives and then re-assayed for radioactivity. If the activity level is at or below background, or meets specific license requirements, it may then be disposed of as nonradioactive waste.

- **Discharge to the Sanitary Sewer**
  Most aqueous liquid radioactive wastes generated by laboratories may be discharged to the sewer if the facility is served by a publicly owned treatment works (POTW). These wastes should not be discharged at facilities served by septic systems.

  The amount of radioactive waste released into the sewerage system is regulated by NRC limits (10 CFR 20.301). Generally, aqueous liquid radioactive waste may be disposed of via the sanitary sewer system, if the following conditions are met:
  - Each approved user of radionuclides must use no more than one “hot” sink for disposing of the liquid waste, unless given an exception by the EHS Section. This is necessary to validate concentration calculations, because federal regulations limit the quantity of radioactive material that can be released into the sewer system. Alternatively, some institutions do not allow users to discharge wastes at individual laboratory sinks. Instead, all wastes are collected in containers, inventoried, and discharged from a central facility.
  - The quantity that can be released to the sewer in any one day cannot exceed the larger of the following:
  - That quantity, if diluted by the average volume of sewerage effluent, that will result in a concentration equal to that

specified in Appendix B, Table 1, Column 2 of 10 CFR Part 20; or

- Ten times the quantity specified in Appendix C of 10 CFR, Part 20.
- The average monthly concentration of radioactive material cannot exceed the limit specified in Appendix B, Table I, Column 2 of 10 CFR Part 20. Applicable state regulations, if any, should also be consulted.
- The total amount of radioactive material released into the sewer system by the laboratory (except for 14C and 3H) cannot exceed one curie per year. For 3H, a maximum of 5 Ci (185 GBq) a year is allowed, while for 14C a maximum of 1 Ci (37 GBq) a year is allowed.
- Liquid waste must be readily soluble or dispersible in water. Any radioactive solids in the waste must be in dispersible form (biodegradable solids).
- Each sink used for radioactive liquid waste disposal should be identified and display the appropriate radiation warning sign.
- Do not exceed the daily limits of radioactive material release posted on each sink by the EHS Section.
- Follow all releases of radioactive waste with copious amounts of water.
- Wastes containing oils, solvents that are in ignitable concentrations or not miscible with water, and other materials regulated as hazardous waste or restricted by the POTW should not be discharged.
- Do not dispose of radioactive material via the sewer system that can be conveniently decayed in storage.
- Excreta from persons undergoing medical diagnosis or therapy with radioactive material are exempt from any of these limitations.

Table 7: Recommended Management Methods for Radioactive Waste Streams Commonly Generated by Laboratories

	Disposal Methods		
	Preferred	First Alternative	Second Alternative
^{131}I waste from iodinations in laboratories, waste from therapy doses administered to patients (amount: less than 10 mCi (0.37 GBq) of ^{131}I)			
Solids	Decay, then	Incineration	
Liquids	Decay, then	Sewer	
Imaging agents with half-life greater than 60 days in diagnostic amounts			
Solids	Decay	Incineration	
Liquids	Decay	Sewer	
Imaging (^{99m}Tc) in diagnostic amounts			
Solids	Decay, then	Incineration	
Liquids	Decay, then	Sewer	
Lead shielding			
Solids	Reuse	Decontaminate, then return to supplier*	Recycle
Liquid scintillation vials, nonignitable counting fluid, with only ^{14}C or ^{3}H at less than 0.05 µCi/g			
Solids	Recycle off-site	Incineration	Normal trash
Liquids	Recycle	Sewer	Incineration
Liquid scintillation vials, nonignitable counting fluid, with ^{14}C and ^{3}H greater than 0.05 µCi/g, or other radionuclides			
Solids	Decay, then	Recycle	
Liquids	Decay, then	Recycle	Sewer
Radioimmunoassay (RIA) Kits (^{125}I)			
Solids	Normal trash†	Decay, then	Incineration
Radionuclides used for *in vivo* kinetic studies			
Solids	Normal trash	Incineration	
Liquids	Sewer		
Source vials, empty			
Solids	Dispose as solid LLW		
Source vials, not empty			
Solids	Return to vendor	Case by case determination	Dispose as Solid LLW
Liquids	Return to vendor	Case by case determination	
^{133}Xenon			
Gas	Charcoal trap	Decay in hood	Release into atmosphere

* Lead decontamination solutions may require treatment as mixed waste.
†Only with general license 10 CFR 31.

- **Medical/Biohazard:**
  Medical waste classification and disposal requirements are defined in UAE Law (4)/1998.Regarding Medical Waste Disposal in the Emirate of Abu Dhabi. The law defines the following groups of medical waste:
  - **Group A** – Anatomical or pathological waste, waste contaminated with human blood or other body fluids, excreta, vomit, human tissue, wastes from contagious diseases, dirty bandages, bedsheets, animal remains and all other materials on which animal lay or cloth or used by animal whether contaminated or not and mortuary wastes.
  - **Group B** – Sharps, usually syringes and needles, surgical tools, different medicine and medical equipment vessels,

broken glass and all other sharp equipment, tools and materials.

- **Group C** – Blood, tissue and microbial cultures and microbiology laboratory waste, carcasses of inoculated lab animals, stools from cholera patient or body fluid of highly infectious diseases, and mortuary waste not specified under Group A.
- **Group D** – Pharmaceutical and chemical waste to which medical specifications apply.
- **Group E** – Deposable linings used for patient beds, caps of bottles for receiving and storing blood, urine, urine diapers, bags or vessels used for receiving stomach waste and similar wastes.
- **Group F** – Waste resulting from treatment with radioactive materials and wastes resulting from all operations related to radioactive materials.

- **Laboratory Infectious Waste:**
  Infectious waste is broadly defined as waste capable of producing infectious disease. This functional definition is based on four factors necessary for disease transmission: the presence of pathogens capable of producing disease, pathogens of sufficient virulence to produce disease, a portal of entry, and a susceptible host.

  Infectious Waste Management includes the following six categories of infectious waste:

Table 8: Categories of Infectious Waste

Category	Examples
(1) Cultures and stocks of infectious agents and associated Biologicals	• Specimens from medical and pathology laboratories • Cultures and stocks of infectious agents from clinical, research, and industrial laboratories; disposable culture dishes; and devices used to transfer, inoculate, and mix cultures • Discarded live and attenuated vaccines
(2) Pathological waste	• Wet tissues, organs, body parts, blood, and body fluids removed during surgery, autopsy, and biopsy
(3) Human blood and blood products	• Waste blood, serum, plasma, and other blood components
(4) Contaminated sharps	• Contaminated hypodermic needles, syringes, scalpel blades, Pasteur pipettes, broken glass,

	used cover slips, and used glass slides
(5) Contaminated animal waste	• Contaminated animal carcasses, body parts, and bedding of animals known to have been exposed to infectious agents
(6) Isolation wastes	• Blood, excretion, exudates, and secretions from patients with highly communicable diseases

a) Handling Infectious Waste:

Personal protective equipment, specifically gloves and protective clothing, must be worn when infectious waste is handled. Water-resistant utility gloves, rather than thin latex or vinyl gloves, are recommended for handling bagged infectious waste. Puncture-resistant gloves should be used when handling containers of infectious waste that contain sharps (i.e., needles, blades, and laboratory glass). Waste handlers should wash their hands after removing their gloves. Additional precautions are needed when waste is disposed of through the sanitary sewer.

Infectious waste should not be compacted except in compactors that are specifically designed for medical waste.
To the extent feasible, handling and movement of infectious waste from the site of generation to the treatment equipment or the loading dock should be minimized.

Medical waste that is multihazardous requires special management procedures that must be tailored to the hazards that are present.

b) Source Separation and Waste Segregation:

Implementation of a waste management program requires segregation of the different waste types into designated categories to allow for the safest, most efficient, and most economical treatment or disposal options. Separation of waste should be performed at the point of waste discard by personnel knowledgeable in the origin and hazard potential of the waste (Medical Laboratory).

All infectious waste should be discarded directly into Biohazardous plastic bags or other containers that are clearly identifiable and distinguishable from the general waste stream. All containers used for infectious waste, including sharps containers, must be red in color or labeled with the “Biohazard” symbol in accordance with HAAD regulations.

Waste receptacles that are easily distinguishable and appropriately marked for contaminated and for non-contaminated waste should be readily available in each area where infectious waste is generated.

Careful separation of infectious waste from the general waste stream will reduce extra handling and treatment costs. When non-contaminated material (such as wrappings and opened but unused materials) is kept out of the infectious waste stream, further savings will result.

Once separated from the general waste stream, infectious waste may be further segregated by the intended treatment method. For example, all laboratory cultures should be autoclaved in the laboratory, whereas other treatment methods may be more appropriate for other types of infectious waste.

c) Packaging and Labeling:

Infectious waste must be contained to protect Laboratory workers, students; waste handlers, and the general public from exposure to the waste and from puncture/ abrasion injury and disease transmission.

Infectious waste should be contained from the point of discarded until the material is no longer infectious and does not pose a risk of injury. Consequently, the container should be designed to maintain its integrity throughout handling, storage, movement, and shipping. In some instances, double bagging or placement of the bags in semi-rigid or rigid containers may be necessary.

Selection of packaging materials should take into account the type and volume of waste, its moisture content, handling procedures, and the treatment technique. Solid and semisolid wastes can usually be packaged in plastic bags. Liquid infectious waste should be placed in leak-proof, stoppered bottles/ flasks, or containment tanks. All containers including plastic bags must be securely closed to prevent leakage and spills.

Polyethylene or polypropylene bags must be red/blue/yellow or labeled to denote the infectious waste content (as per HAAD color coded bags). Plastic bags should have sufficient tensile strength to resist rupture from rough handling or overfill conditions. The packaging should be compatible with treatment mechanisms. Specifically, the packaging materials should not thwart the decontamination process.

Waste haulers, disposal agencies, or landfill operators may specify packaging requirements when an infectious waste is to be shipped off-site for treatment or disposal. However, packaging should, at a minimum, be consistent with safety practices discussed in this guideline.

Packaging and color codes for medical waste (reference Schedule 3, Federal Regulation)

- Group A waste: (Anatomical, Pathological waste) Heavy duty polyethylene (gauge 400) red color bags clearly marked with the phrase “Contagious Wastes”.

- Group B Waste: (Sharp Objects) Heavy duty thick polyethylene plastic boxes of yellow color internationally known as “Sharp Object Boxes” clearly marked with the phrase “Sharp Objects”. The boxes for preserving such waste shall be tightly sealed with a cap or lock or any other means so as not be opened and must not be stowed for more than 75% of their capacity.

- Group C Waste: (Laboratory waste)

  - Waste to be sterilized before disposal: All laboratory waste such as bacterial plates, culture plates, gloves, contaminated clothes, sheets, covers and vessels used for handling organic tissues, contaminated blood etc. These wastes shall be placed in blue transparent bags of special material clearly marked up “Medical Waste- to be autoclaved/ sterilized before Disposal” After autoclaving it needs to be packaged in Red bags. These bags need to be tied and should not be packed for more than 65% of their capacity.

  - These groups include all other waste under Category C not included above. These wastes must be placed in medium duty (gauge 200) polyethylene bags of yellow color clearly marked up (Medical waste). These should not be filled more than 65% of their capacity and shall be tied, and stored waiting collection and disposal by incineration.

- Group D waste: (Pharmaceutical & Chemical waste)

  - Pharmaceutical waste: All packed pharmaceutical partially used or expired shall be returned to the pharmacy to their original containers, then stored in bags of polyethylene of medium duty (gauge 300) of yellow color marked up (Medical wastes) . Bags should be tied in their neck and should not be filled more than the capacity and shall be stored until collection and disposal.

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- Poisonous Waste: All cellular or poisonous materials shall be returned to a predetermined point inside the pharmacy in which such materials are prepared or issued. All poisonous cellular medicines and other materials contaminated with cellular poisonous drugs and medicines (except for under skin needles and other sharp objects classified under Group B waste, shall be placed in heavy duty (gauge 400) polyethylene bags of yellow color clearly marked up “Poisonous (Cellular Cytotoxic) Waste-” and such bags should not be filled more than 65% of their capacity and tied in the neck and properly stored until collection and disposal by incineration.

- Chemical waste: All Chemical waste to which medical waste characteristics apply as industrial solvents and other liquid materials used in diagnostics tests in addition to all remaining chemical materials shall be returned to a predetermined point in the pharmacy or the central laboratory store where they shall be tagged according to their respective kinds and types either by using adhesive tags or cards attached. Tags and marks placed on each pack shall indicate its components and hazards. Containers then are put into yellow bags after that shall be stored awaiting collection and disposal.

- Group E waste: All used materials receiving patient secretions and stomach waste and other waste (except patient suffering from contagious disease listed under Group A) shall be placed in medium duty (gauge 300) yellow polyethylene bags. These bags should not be filled more than 65% of their capacity, shall be tied, and tagged as Medical waste of Group E. These bags must be isolated in a separate place from other medical waste (usually cut into small pieces and disinfected before treatment).

- Group F waste (radioactive waste): Must be kept in special security areas allocated for storing this kind of waste until expiry of half of their expected lives stored in special containers , marked and labelled and sent back to the manufacturer.

d) Movement of Infectious Waste Within the Facility:

Waste should be moved in clearly labeled, dedicated, leak-proof carts. These carts must be decontaminated before being used to move other wastes or supplies. Carts used for waste movement should be disinfected frequently. Routes in the university facility should be designed to reduce the

risk of exposure to employees, Students, and visitors. Service elevators, rather than public elevators, should be used for waste movement.

It is essential to maintain the integrity of the packaging during handling, transferring, and loading of containers of infectious waste. Mechanical devices should not be used to transfer or load infectious waste, because rupture of the packaging and spillage of waste may occur. Chutes and dumbwaiters should not be used. Waste should be loaded by hand.

e) Treatment of infectious Waste (Autoclaving):

Autoclaving uses saturated steam under pressure to decontaminate infectious waste. Infectious laboratory waste, instruments, and glassware can be decontaminated in a gravity displacement or prevacuum autoclave. The effectiveness of autoclaving is determined by various factors, including:

- Amount of waste;
- Physical characteristics of the waste load (including size, density, and organic content);
- Use and number of autoclavable bags (self-venting bags or an acceptable alternative);
- Type of container (metal or polypropylene);
- Steam penetration throughout the entire waste load and direct contact with microorganisms;
- Exposure time; and
- Exposure temperature.

In addition, all autoclaves must be monitored at least weekly for proper functioning using a biological indicator (e.g., Bacillus stearothermophilus spores) or its chemical equivalent. Laboratory autoclaves normally operate at a temperature of 250 °F (121 °C), a pressure of 15 psi, and a cycle time of 30 minutes. Commercial autoclaves operate at a higher temperature of 275 °F (135 °C), a pressure of 60 to 75 psi, and a cycle time of one hour. Steam sterilization can be reliable if extended exposure periods are used and conditions are optimized for appropriate heat transfer throughout the waste load.

f) Management of Contaminated Sharps

The sharps category consists of all items that can easily penetrate plastic bags and many other containers. This includes not only the obvious needles and blades, but also any object that is readily breakable. The

sharps category includes discarded hypodermic needles; syringes (with or without the attached needle); scalpel blades; lancets; capillary tubes; micropipettes; slides and cover slips; and wooden applicator sticks, as well as broken laboratory glassware (e.g., pipettes, blood tubes, and culture dishes).

From the management aspect, sharps are of two types: contaminated and non-contaminated. Contaminated sharps are those that were used or may have come in contact with infectious agents. Non-contaminated sharps consist of items such as broken glassware and shards of hard plastic that were not in contact with infectious agents.

Sharps should be discarded directly into containers that are readily accessible at laboratory situations to ensure utilization and to minimize the distance that used sharps must be carried for disposal.

KU requires sharps collection containers to be closable, puncture-resistant, leakproof on the sides and bottom, and labeled or color-coded. Many designs of puncture-resistant sharps containers are available; choose one that is most appropriate and works best for the waste type. Sharps should not be handled directly; loose sharps should be handled with tongs or by some other mechanical means. Needles should not be recapped, bent, or sheared, and they should be placed with syringes intact into the sharps containers.

All sharps should be continued from the point of discarded until they are no longer hazardous. They should be placed into rigid, puncture-resistant containers immediately upon discard. The containers can be of metal, wood, or rigid plastic. Polystyrene, nonrigid plastic, paper, and cardboard containers are not acceptable.
To prevent needlestick injuries during handling and disposal, the following precautions must be taken:

- Discard needles directly into rigid, puncture-resistant containers.
- Do not re-sheath needles by hand. Available devices include protective guards that permanently enclose the needle after use and needle containers with integral devices that facilitate removing needles from needle holders.
- Do not remove used needles from disposable syringes. A one-handed recapping technique should be used whenever the needle must be removed.

- Do not shear, bend, break, or clip needles.
- Avoid manipulating needles as much as possible.
- Do not place needles in Biohazard bags unless they are already in a rigid, impervious, puncture resistant container.
- Do not discard needles with general waste.
- **Specific Wastes:** Containers/Drums/Compressed Gas Cylinders

1. Any drum or containers that held any acutely hazardous substance will be considered hazardous regardless of the quantity of substance remaining in the drum or container. See Table No. 2 – substances marked with an asterisk (*)
2. Any container or cylinder awaiting disposal, or being discarded and contains a hazardous compressed gas with a pressure greater than (>) atmospheric pressure, must be classified as hazardous waste.
3. Electrical equipment (non-medical) must therefore be disposed of after Facilities Management Department declaration and approval. No electrical equipment must be disposed of via the general waste stream, but must be segregated and removed from the site by a registered contractor.
4. The definition of electrical equipment is any item of equipment that uses electricity to function, for example:
   - Televisions
   - Toasters
   - Mobile phones
   - Microwaves
   - Fridges
   - Freezers, etc.
5. If you have any electrical equipment that has been condemned, contact the Facilities Management Department who will arrange tag system for its disposal in line with this procedure. Please ensure that the item is marked “condemned - for disposal” to allow the porters to easily identify the item.
6. Medical equipment should still be decontaminated and disposed of via the Facilities Management Department as per Trust protocol.
7. All IT equipment must be decommissioned by the IT Department to preserve confidentiality, therefore any IT equipment (computers,

VDU's, printers, handheld units, laptops etc.), must be sent to IT department who will arrange for their appropriate disposal.

8. Any items of furniture should be marked "redundant furniture for disposal" and stored in a safe manner. Contact Facilities Management Department, who will collect the item, and store it in the main waste compound to await collection by the appropriate contractor,
9. Any construction or demolition waste will be removed from site by the contractor involved. Facilities Directorate will ensure that the appropriate license's or documentation is current.
10. A safe storage area must be designated while taking into account the possibility of;
    - Accidental spillage or leakage.
    - Contamination of soils and groundwater.
    - Corrosion or wear of containers.
    - Loss of the container's integrity from accidental collisions.
    - Deterioration due to UV and weather exposure.
11. The waste containers are clearly marked. The following waste management color coding system is being implemented to help in waste type Identification and to minimize waste mixing.

Table 9: Color coding of waste collection bins internal KU building and premises:

Waste Type	Waste Collection Bin Color
General Waste (Non- Hazardous waste for landfill / incineration)	**Black**
Paper / Carton	**Blue**
Recyclable (Cans)	**Yellow**
Recyclable (Plastic)	**Green**
Biohazard (autoclaved)	Red
Electrical, Electronics & Computer consumable	To be collected segregated and disposed Properly by special contractor
Battery Waste	To be collected segregated and disposed Properly by special contractor
Printer Cartridge	To be collected segregated and

Waste Type	Waste Collection Bin Color
	finally collected by the supplier.

Table 10: Color coding of waste collection skips / containers:

Waste Type	Waste Collection Bin Color
Recyclable solid Material, scrap waste (Plastic, Glass, Paper, Plastic, Metal	**Black**
Biodegradable Waste/ Medical Waste	**Red**
Solid Medical Waste	**Blue**
Construction & Demolition Waste / non – hazardous waste	**Green**

The identification of different waste streams is based on:

- Standard Operating Procedure (S.O.P.) for the Approval of Registering Waste Containers & Skips – The Center of Waste Management-Abu Dhabi.
- Law No (21) for the year 2005 on Waste Management in the Emirate of Abu Dhabi.
- Classification of hazardous substances, hazardous wastes, radioactive and medical wastes stipulated in Regulation Concerning Handling of Hazardous Substances, Hazardous wastes and Medical Wastes under Federal Law No. (24) of 1999.
- Decree No 17 of 2008 establishing the Centre of Waste Management – Abu Dhabi.
- Decree 2C24 of 2009 for Tariff System of the Waste in the Emirates of Abu Dhabi.
- Center of Waste Management Board of Directors Decree No 1/2010 for the Waste Tracking System.
- Definition of wastes as stipulated in Abu Dhabi Law No. (21) of 2005.
- Hazardous Materials Management doc. No. KU – FD – EHS – MPM – 02.

5.6 Waste Container Requirements for All Generators:

Requirements for chemical waste containers at the point of generation as well as during handling, accumulation, and on-site storage include:

- Collecting waste in a container that is compatible with the waste type (i.e., chemically resistant plastics for solvents, glass for mineral acids);
- Using a “Hazardous Waste” label or marking, including the chemical name (e.g., “waste Xylene”); and
- Making sure the container is tightly capped at all times except when adding or disposing of waste.

5.7 Waste Management Considerations:

For each category of laboratory waste, the following issues need to be addressed:

- Source reduction, waste minimization, and pollution prevention;
- Characteristics and examples;
- Handling (including segregation, engineering controls, packaging, labeling, containment, and use of Personal protective equipment);
- Storage;
- Accumulation at the point of generation;
- Treatment (on-site);
- Transportation;
- Disposal;
- Waste minimization and hazard abatement; and
- Regulations and exemptions (as appropriate).

5.8 Waste Generators:

- All efforts are being dedicated towards minimizing waste generation or preventing it at the source.
- The Department/ Section or person that generates waste:
  - Classify waste as per this procedure.
  - Segregate non-hazardous waste to be recycled into the appropriate recycle bins.
  - Ensure all wastes are stored correctly to minimize adverse impact to the environment and human health.
  - Ensure that incompatible wastes are not mixed.
  - Ensure that Waste Disposal Contractor / Environmental Service Provider, which KU uses, have a valid permit from the Environment Agency of Abu Dhabi / Centre of Waste Management–Abu Dhabi (CWM – AD).

- Ensure that the waste management facility, which KU uses, is permitted to receive the classified waste.

- The waste generator remains the 'owner' of the waste and responsible for correct handling of the waste in accordance with this procedure until it reaches the waste management facility.
- The waste generator is required to keep receipts or certificates from the final destination of the waste for a period of 5 years.
- The generator of waste is required to keep records (Waste Declaration Form to be completed and send to EHS Section Monthly) or logs of waste produced, generation process, and the amounts generated and transported to the waste treatment / storage facility.
- The generator of hazardous wastes.
  - Be responsible for classifying the waste.
  - Determine the:
    1. Potentially hazardous properties of the waste, and
    2. Constituents of the waste that give rise to those potentially hazardous properties.
  - Ensure hazardous waste is stored and managed according to the identified hazards to minimize risk to human health and the environment; and
  - Complete relevant sections of the Manifest for Transporting Toxic and Hazardous Wastes.

5.9 Accumulation and Storage Requirements

KU uses the term "accumulation" to mean the on-site accumulation of hazardous waste prior to shipment to an offset, permitted treatment, storage, or disposal facility (TSDF).

Large- and small-quantity generators may accumulate hazardous waste only in locations that meet requirements for accumulation facilities, or at or near the point of waste generation—which is called "satellite accumulation." There is no time limit for storage of hazardous waste. However, the generators in this category may store no more than 1,000 kg at any time.

5.10 Waste storage:

The storage sites are planned in such a way as to minimize inadvertent access by the untrained or unauthorized. Safe storage requires engineering controls appropriate to the type of waste.

While it is economically desirable to reduce the number of waste shipments, storage times must be weighed against the costs and hazards of storage. For example, radioactive materials with a short half-life may be allowed to decay in storage until they are safe for disposal.

Infectious waste autoclaved and stored in a designated A/C-ventilated, lockable and closed room (maximum air temperature +15 °C) with access limited to authorized personnel only. All infectious waste must be disposed of within 24 hours.

Chemical wastes shall be segregated and returned to the supplier if possible or central laboratory where they shall be tagged according to their type. Tags and marks placed on each pack shall indicate its components and hazards. Individual containers shall then be placed in yellow bags or containers and stored in a locked room until collection and disposal.

5.11 Waste Transfer / Disposal

The waste shall only be transferred / disposed by Waste Disposal Contractor / Environmental Service Provider, who has a valid permit from the Environment Agency of Abu Dhabi / Centre of Waste Management–Abu Dhabi with proven experience and infrastructure for the transport and / or disposal of wastes. These contracting companies have to verify that the transportation, treatment and final disposal of the wastes complies with the requirements of the UAE Statutory requirements. Waste management practices implemented in the laboratory, are detailed out in Laboratory Handbook.

When plastic garbage is not separated from other garbage, the mixture must be treated as if it were all plastic. The residue from plastic Incineration is still considered plastic and thus cannot be discharged overboard. Packaging from hazardous / harmful substances shall be returned to the suppliers. When garbage is mixed with other harmful substances having different disposal or discharge requirements, the more stringent disposal requirements will apply.

The waste management producer will identify the recyclable waste within his / her area of operations. For example:

- Large Ni-Cad Batteries.
- Lead acid batteries.
- Some large wet type lithium batteries.

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- Paper waste.
- Plastic.
- Scrap metal.
- Cans.

The recyclable items must be suitably packed in separate containers, and forwarded to the warehouse. An appropriate packaging must be used to ensure their safe transportation. When sent from the work site, the containers with recyclable waste must be clearly marked to indicate;

- Work site where the waste originated.
- Content inside the container.

When sufficient amount of a type of recyclable waste is accumulated, it will be sent to a recycling facility.

5.12 Waste Management Facility:

KU uses a waste management facility has a valid permit to operate, issued by the Environment Agency of Abu Dhabi / Centre of Waste Management – Abu Dhabi Emission rate standards according to the Federal Law, Local Law and AD EHSMS are met.

5.13 Monitoring and Reporting of Generated Waste:

KU periodically monitors the effectiveness of this waste management procedure and update as required.
KU submits regular reports on the amounts of waste managed to the Environment Agency of Abu Dhabi / Abu Dhabi EHS Centre as applicable.

5.14 Treatment and Disposal Options:

Perhaps the most important decision pertaining to hazardous waste management is the selection of treatment and/or disposal methods. These decisions should be based on a consistent strategy that takes into account regulatory requirements, future liability, occupational and environmental risk, institutional self-reliance, cost, and other values important to the institution. Some of these methods may also be considered waste minimization.

- On-**Site Recycling:**

Solvent recycling by distillation is a useful way to recover spent solvents if the solvent can be recycled to a purity that is equal to or better than that of the purchased original, and if the volume of solvent generated makes it

economically feasible. Proper segregation of the spent solvents will make the process more efficient.
KU considers recycling as a means of waste reduction to be the second preferred waste management alternative to their waste management hierarchy. State that material is "recycled" if it is used, reused, or reclaimed.

- **Disposal Via the Sanitary Sewer:**

Disposal of hazardous waste into the sanitary sewer is not allowed, except in limited situations. Following are the criteria for disposal of hazardous waste in the sanitary sewer:

- The hazardous waste must be mixed with domestic wastewater and pass through a sewer system to a publicly owned treatment works for treatment.
- The waste type, amount, and concentration must meet the regulations.

- **Disposal in the Normal Trash:**

Disposal of any amount of hazardous waste in the normal trash or as a solid waste is strictly forbidden.

Off-site disposal of hazardous waste is only allowed at facilities that have CWM – AD hazardous waste permit.

- **Waste Minimization:**

Waste minimization is an essential part of any waste management planning and it will be considered in future program planning. Waste minimization activities are described in Table 11. Waste minimization methods also include some of the methods described in previous sections.

Table 11: Example of Waste Minimizing Activities:

Method	Example
Source separation	• The use of multiple waste collection containers to keep waste types separate, so that each waste type can be optimally managed. • In offices, separate receptacles for recyclable paper.
Waste segregation	• Keeping waste types segregated during collection, storage, transport, treatment, and disposal. • Keeping needle boxes out of red bags.
Source Reduction	• Require purchases to be in small, manageable quantities. This reduces the potential for future disposal costs of used and, more importantly, unused hazardous materials. • Use nonhazardous or less hazardous materials whenever possible. For example, there are substitutes available for xylene in the Molecular laboratory; formalin

	in histology and from surgical specimens; and mercuric chloride salts used in some fixation procedures. Use a scintillation cocktail that is not aromatic hydrocarbon-based. Each laboratory must determine if these products work satisfactorily.
Redistribution	• Identify surplus or unwanted laboratory chemicals, and relocate them to laboratories that can use them.
Recycling	• Solvents like xylene, xylene substitutes, and alcohols can be recycled and reused in the process from which they were generated. Formalin, a fixative, can also be recycled. • Many facilities have set up recycling programs for aluminum cans, paper, and other solid waste items.
Process changes	• Substitute nonchromium cleaners for chromic acid. • Use miniaturized testing procedures to reduce the quantity of hazardous materials needed. • Substitute ozone-friendlier cryogenic sprays. • Use xylene substitutes in tissue processing, staining, and cover slipping.
Recovery	• Silver recovery units. • Heat recovery using waste solvents for fuel. This requires proper attachments to the facility fuel burner and may require permits in some areas. Be sure to check additional federal and state requirements.
Hazard reduction	• A treatment or procedure that reduces the hazard of the waste, making it easier or less expensive to dispose of. • When allowed, neutralize waste acids so that they can be disposed of in the sanitary sewer.

5.15 Contingency Planning and Employee Training:

Train employees to properly handle the waste and follow emergency procedures, as appropriate to their responsibilities. CWM – AD requires annual training for employees who handle hazardous chemical wastes of large- and small-quantity generators.

For first responders and spill-response teams, OSHA training standards apply. Spills are common chemical emergencies in laboratories. Spill cleanup procedures are specific to the chemical.

5.16 Waste Records:

Whenever waste is transferred, a “Waste Transfer Form” is completed, signed and kept by the parties involved in the waste generation or transfer and any subsequent processing. And these records are maintained for 5 years.

Waste Transfer Form will be completed when the Waste Handling Contractor removes waste from KU and submits the waste management record to KU.

6. Reference Standards:

6.1 Abu Dhabi Occupational Safety & Health System Framework (OSHAD-SF) - CoP-54 Waste Management.

6.2 Law No. (21) of 2005 Concerning Waste Management in Abu Dhabi Emirate.
Clinical Laboratory Waste Management; Approved Guideline—Second Edition, GP5-A2, Vol. 22 No. 3

6.3 ISO 45001:2018 Occupational health and safety management systems – Requirements.

6.4 ISO 14001:2015 Environmental Management System – Requirements.

7. Annexure:

7.1 Nil

8. Records / Documents / Formats:

8.1 Waste Declaration Form (Internal Use): KU – FD – EHS – FOR – 70.

8.2 Waste Transfer Form: KU – FD – EHS – FOR – 71.

8.3 Waste Inventory Register Form: KU – FD – EHS – FOR – 72.

8.4 Waste Inventory Register doc. No.: KU – FD – EHS – REG – 22.

8.5 Lab waste disposal request Form: KU – FD – EHS – FOR – 149.

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Record of Reviews & Revisions

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		PAGE NO	COMMENT		
01	07/06/2018	All	Change of logo	Roberto M.	Belal I.
		Throughout	Change from KUSTAR to KU	Roberto M.	Belal I.
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02	05/01/2020	24-25	Packaging and color codes for medical waste	Navaz K.	Belal I.

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